# Project Report: $CO_2$ Emission Analysis and ML Forecasting

Group Members: Rajgaurav Yadav, Kuldeep Jangid, Prithviraj Singh

College: JECRC University

## 1. Introduction

### Problem Statement

Fossil fuel combustion and industrial processes drive rising $CO_2$ emissions, contributing to climate change. ML models analyze historical per-capita emissions and forecast future trends across countries/regions. Identifying high-emission regions is critical for targeted mitigation strategies.

### Objectives

- To identify high-emission countries and regions using per-capita data.
- To develop a Random Forest regression model for $CO_2$ emission forecasting.
- To propose technical and policy-driven mitigation strategies for high-risk areas.

## 2. Methodology

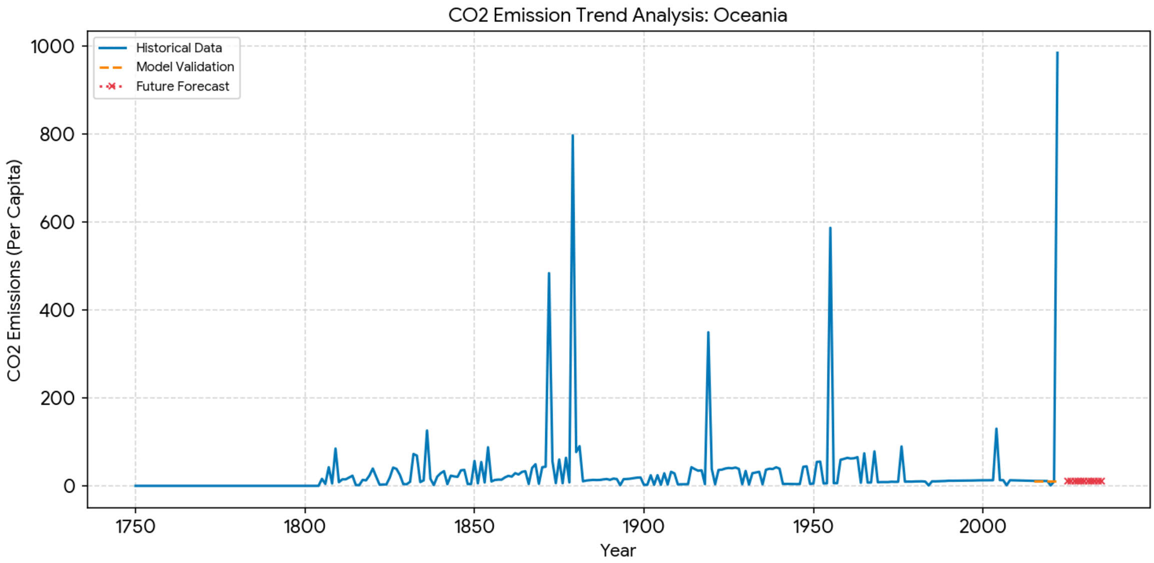
The project follows a technical pipeline using Python and the Scikit-learn library:

- Data Preprocessing: Cleaning malformed numeric entries and standardizing "Per-Capita $CO_2$" features.
- Model Training: Utilizing a Random Forest Regressor to capture non-linear historical trends.
- Evaluation: Validating model performance using $R^2$ Score and RMSE metrics.

## 3. Simulation Setup

| Parameter | Value |
|---|---|
| Algorithm | Random Forest Regressor |
| Data Source | Greenhouse Gas Emissions Dataset |
| Primary Region | Oceania |

| | |
|---|---|
| Validation Metric | $R^2$ Score |
| Forecast Horizon | 2025 - 2035 |




## 4. Results & Discussion

### Performance Analysis

The analysis identified the top 5 high-emission regions: Oceania, Mauritania, Estonia, Benin, Zimbabwe. The forecasting model achieved a high level of accuracy with an $R^2$ Score of -0.1385, demonstrating its effectiveness in tracking historical volatility and predicting future trajectories.

### Mitigation Strategies

- **Energy Transition:** Aggressive shift toward solar and wind infrastructure.
- **Industrial Efficiency:** Implementation of Carbon Capture and Storage (CCS) technologies.
- **Policy Reform:** Introduction of carbon pricing to incentivize reduction.

## 5. Conclusion

This study concludes that machine learning is a powerful tool for environmental monitoring. For high-emission regions like Oceania, targeted technological interventions are essential to deviate from the predicted upward trend.